



Lecture 18: Toggling Classes

INTERVIEW QUESTION

[> CORE](#)

How do you apply and toggle classes conditionally in Svelte?

You want a `disabled` class when the button is disabled, an `active` class when it's pressed. How do you toggle classes based on state?

User interfaces are highly dynamic. When a user interacts with a component, it often needs to visually respond by changing its appearance, which usually requires adding or removing CSS classes dynamically.

The Imperative Class Problem

Imagine you are building a custom `<ToggleSwitch>` component for a dark mode setting. When the state variable `isDark` becomes true, you want the switch to turn blue by applying the CSS class `active`.

In older vanilla JavaScript patterns, you would write imperative code to hunt down the DOM element and manually manipulate its class list. You would write

```
document.getElementById('switch').classList.add('active')
```

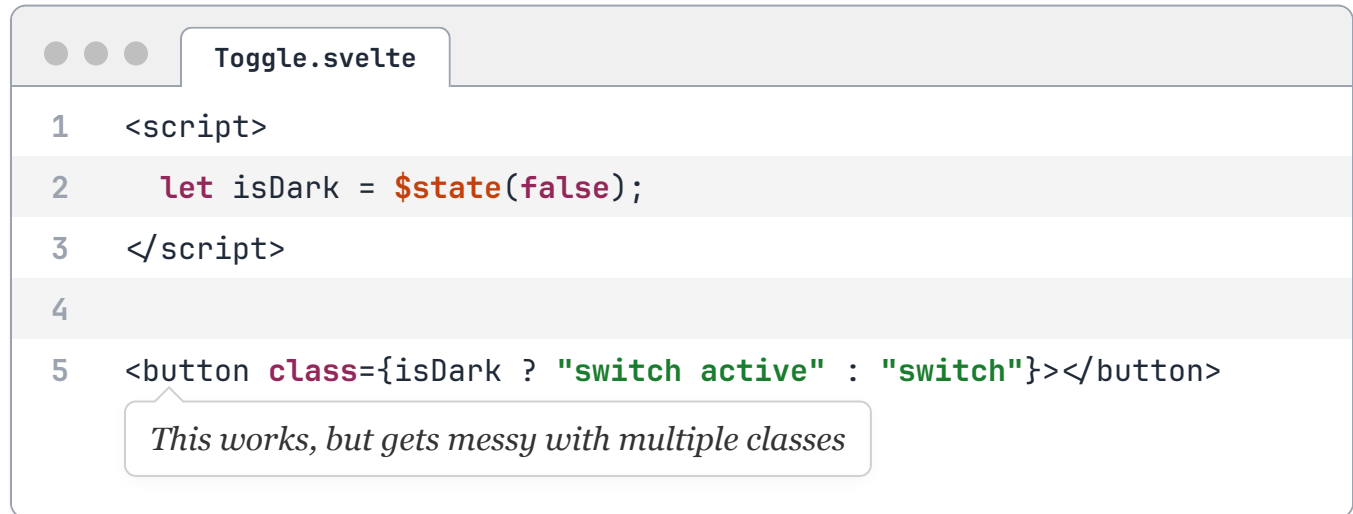
when the setting turned on, and

```
classList.remove('active')
```

when it turned off. This constant, manual synchronization between your JavaScript state and your DOM classes is tedious, prone to typos, and extremely easy to accidentally leave out of sync, resulting in a UI that lies to the user about its true state.

Dynamic Class Binding

Because Svelte's curly braces can evaluate any JavaScript expression, you can theoretically construct a class string dynamically using a ternary operator.



The screenshot shows a code editor window titled "Toggle.svelte". The code is as follows:

```
1 <script>
2   let isDark = $state(false);
3 </script>
4
5 <button class={isDark ? "switch active" : "switch"}></button>
```

A callout box points to the `class` attribute in line 5, containing the text: *This works, but gets messy with multiple classes*

While this works perfectly for a single condition, it quickly devolves into an unreadable mess if you need to toggle three or four different classes simultaneously.

The `class:` Directive

To solve this, Svelte provides a specialized, elegant shorthand called the `class:` directive. It allows you to toggle a specific CSS class on or off based entirely on a boolean condition.

The syntax is `class:name={condition}`. If the condition evaluates to true, Svelte adds the class. If it evaluates to false, Svelte removes it.



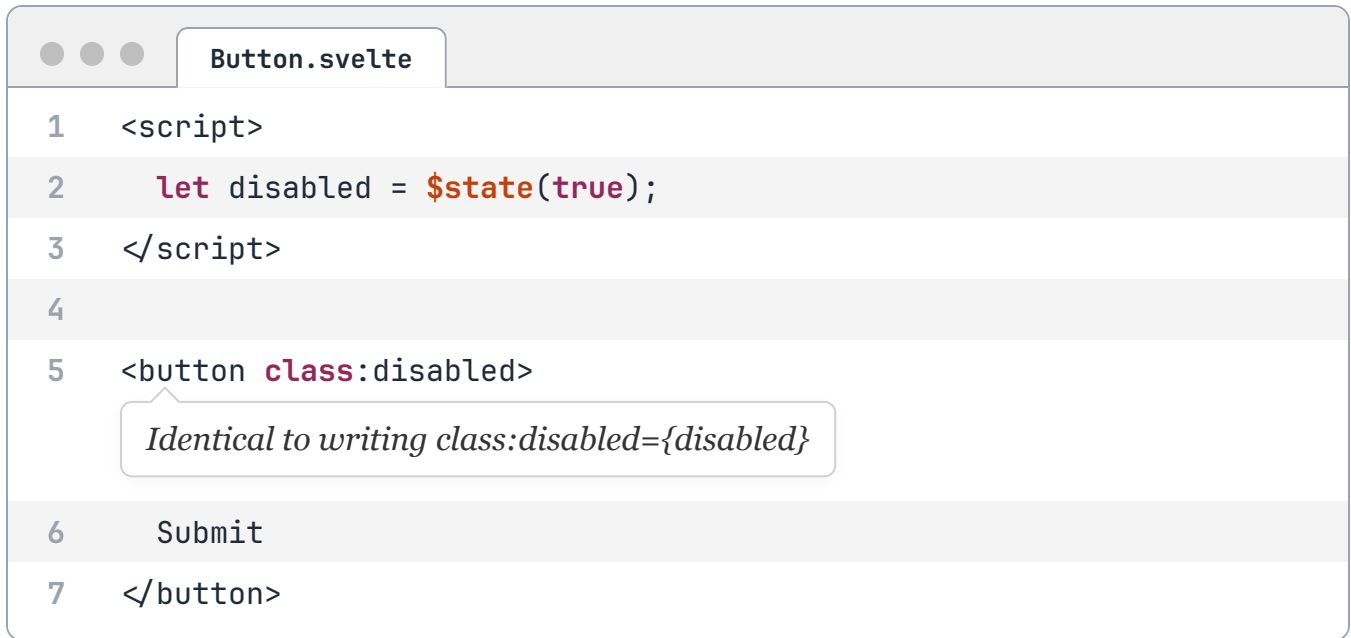
```
1 <script>
2   let isDark = $state(false);
3 </script>
4
5 <button
6   class="switch"
7   class:active={isDark}
8 >
9   Toggle
10 </button>
```

Adds 'active' only when isDark is true

This directive does not overwrite your static `class="switch"`. Svelte intelligently merges them together. If `isDark` is true, the browser receives `class="switch active"`.

The Name-Matching Shorthand

Just like the shorthand for regular attributes, Svelte provides a shortcut for the `class:` directive. If the name of the CSS class you want to toggle is exactly identical to the name of the variable controlling it, you can drop the `=`.



```
1  <script>
2    let disabled = $state(true);
3  </script>
4
5  <button class:disabled>
6    Submit
7  </button>
```

Identical to writing `class:disabled={disabled}`

KEY TAKEAWAY

The `class:` directive is evaluated reactively. You never have to manually add or remove the class using JavaScript; the Svelte compiler handles the DOM updates automatically whenever the underlying `$state` changes.

Summary

To visually adapt your UI based on reactive state, Svelte provides the `class:` directive. Instead of manually pushing strings into a `classList` array, you declare a declarative binding using `class:name={condition}`. Svelte automatically adds or removes the specified class whenever the condition changes, seamlessly merging it with any static classes you have defined. When the class name and the variable name match perfectly, you can use the ultra-clean `class:name` shorthand.

	VANILLA JS (Imperative)	SVELTE (Declarative)
Adding	<code>el.classList.add('active')</code>	<code>class:active={true}</code>
Removing	<code>el.classList.remove('active')</code>	<code>class:active={false}</code>
Shorthand	N/A	<code>class:active</code>